Task 2 - Developing the solution

The solution

Indicative content and marker guidance		
Functionality: • Must demonstrate the use of two different languages, JS and PHP or SQL. Python and SQL. • The techniques used are appropriate and demonstrate a level of technical skill • Maybe complex data model in database interlinking more than one table • Recursive algorithms – reminder that if used in PHP here must be a mechanism (IF statement, etc.) that stops the recursion after the desired result has been found • Server-side scripting using request and response objects and server-side extensions for a complex client-server model • Web service APIs and parsing JSON/XML to service a complex client-server model • Use of MySQL, the use of DML and DDL • Various outcomes which may be basic in terms of very linear structure to more logically and efficient approach. The use of function, procedures and Classes • The solution will be well structured and modular in nature – clearly see subsections this may be separated modules •	**Code organisation:** • For top marks should avoid multiple pages of nested if clauses and for loops with a lot of copy-pasted procedural code • Clear and meaningful indentation • Code should consist of pieces of logic, classes or objects, with proper structure. • comments should be used where ever possible to help explain the logic • good use of local variables and minimal use of global variables • use of constants • well-designed interface • consistent style throughout • defensive programming • good exception handling	**User Experience considerations:** • Useful – does it meet the needs of the user? Are outputs etc. accurate • Usable – is it easy to use / intuitive? • Desirable – is it aesthetically pleasing? Does it represent a brand? Consistency. • Findable – is it easy to navigate around and find the required information/features • Accessible – what accessibility features have been included? Are these well implemented? Are they sufficient? • Credible – is the information credible?

Assessment focus	Band 0	Band 1	Band 2	Band 3	Band 4
	0	1-2	3-4	3-5	6-8
Functionality	No rewardable material	The prototype implements code in a single language with some functionality but the code lacks efficiency and some major errors persist. Uses some precise logic and programming structures which would result in some correct outcomes	The prototype implements code with some functionality in at least two different languages but the code lacks efficiency and some major errors persist. Uses sufficient precise logic and programming structures which would result in adequate correct outcomes.	The prototype implements mostly efficient functional code in at least two different languages, but some minor errors still persist. Uses mostly precise logic and programming structures which would result in mostly correct outcomes.	The prototype implements consistently efficient functional code in at least two different languages. Uses precise logic and programming structures throughout which would result in consistently correct outcomes.

		1-2	3-4	3-5	6-8
Code organisation		Code is maintainable by a third party, but would present significant difficulties through the use of: • inconsistent naming conventions. • limited logical organisation • limited informative commenting.	Code is maintainable by a third party, but would present some difficulties through the use of: • somewhat appropriate naming conventions. • Some logical organisation • some informative commenting.	Code is maintainable by a third party, and would present only a few minor difficulties through the use of: • mostly appropriate naming conventions. • mostly logical organisation • mostly informative commenting.	Code is easily maintainable by a third party through the use of consistently appropriate: • naming conventions. • logical organisation • informative commenting.

		1-2	3-4	3-5	6-8
User experience		Basic user experience is provided through limited effective use of: • input handling • user guidance and error messages • outputs The solution is partially robust and effectively handles some common errors	Adequate user experience is provided through somewhat effective use of: • input handling • user guidance and error messages • outputs The solution is adequately robust and effectively handles sufficient common and unexpected errors	Good user experience is provided through mostly effective use of: • input handling • user guidance and error messages • outputs The solution is largely robust and effectively handles most common and unexpected errors	Excellent user experience is provided through consistently effective use of: • input handling • user guidance and error messages • outputs The solution is fully robust and effectively handles common and unexpected errors

		1-2	3-4	5-6	
Legal and regulatory guidelines and Standards		Some effective application of standards and guidelines in relation to: ● accessibility ● compatibility ● legal and ethical considerations Some effective application of procedures and security controls to ensure confidentiality, integrity and availability.	Mostly effective application of standards and guidelines in relation to: ● accessibility ● compatibility ● legal and ethical considerations A mostly effective application of procedures and security controls to ensure confidentiality, integrity and availability.	Consistent and effective application of standards and guidelines in relation to: ● accessibility ● compatibility ● legal and ethical considerations Thoroughly effective application of procedures and security controls to ensure confidentiality, integrity and availability.	